

Authentication and authorisation

Project: Grounded Learning Service

1. Your mission

Your learning target

I can explain the mechanism and terminology of authentication and authorisation.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of authentication and authorisation

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Configuration and secrets. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain authentication and authorisation using the deciding step in this fictional service.

2. Notice the evidence

Authentication identifies an actor; authorisation decides whether that actor may perform a particular operation on a particular resource.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Authentication and authorisation — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: A student session submits role=teacher in request data and asks for a guide.

Check or calculation: Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Conclusion and limit: The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner

should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Use a verified curator permitted to edit C.

Prediction: The authorisation gate allows the edit, subject to other declared checks.

Reason: The trusted role now matches this operation and object.

5. Words that unlock the idea

Word · Meaning

Word: Authentication

Meaning: Checking an actor identity or session

Word: Authorisation

Meaning: Checking permission for an operation/resource

Word: Trusted role

Meaning: A role established by the service rather than user-claimed data

Use two terms to explain a step in this week's example.

6. Follow a worked example

A student session submits role=teacher in request data and asks for a guide.

The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
def may_read(role, resource_role):  
    if role not in ('student', 'teacher') or resource_role not in  
    ('student', 'teacher'):  
        return False  
    return role == 'teacher' or resource_role == 'student'  
  
print(may_read("student", "teacher"))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Authentication and authorisation

A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.

The facts we will use

Trusted policy grants volunteers read access to C; only curators may edit it.

AI ACADEMY / AI-6/5

Authentication and authorisation

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

**1. Identify Jo**
Authentication establishes Jo's identity and volunteer role.



**2. Check the action**
Editing C requires curator authority.



**3. Deny the edit**
Keep C unchanged; the authenticated volunteer lacks this permission.

CHANGE ONE THING
Use a verified curator permitted to edit C.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Identify Jo

What happens and why: Authentication establishes Jo’s identity and volunteer role.

Step: 2. Check the action

What happens and why: Editing C requires curator authority.

Step: 3. Deny the edit

What happens and why: Keep C unchanged; the authenticated volunteer lacks this permission.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Authentication establishes identity. Authorisation compares that identity with the requested action and resource.

What this example does not establish

A role written in an ordinary request is not a trusted authentication claim.

7. Find and repair a mistake

Successful authentication does not grant every resource or operation.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Successful authentication does not grant every resource or operation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Authentication and authorisation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Authentication and authorisation. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Explain why Jo's valid volunteer login does not permit editing C. Which trusted role change allows the action, and why is a role written in the request insufficient?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

```
def may_read(role, resource_role):  
    if role not in ('student', 'teacher') or resource_role not in  
    ('student', 'teacher'):  
        return False  
    return role == 'teacher' or resource_role == 'student'  
  
print(may_read("student", "teacher"))
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Successful authentication does not grant every resource or operation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse

W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled authentication and authorisation evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of authentication and authorisation. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Authentication and authorisation — evidence trace

Input and scope · Check or calculation · Conclusion and limit

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cannot change authenticated authority.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.